CRNN Architecture and Feature Extraction

Kaushal Vaid
Roll Number: 24f2000681@ds.study.iitm.ac.in

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Overview

This report details the fourth milestone phase of data processing and model building for the "Messy Mashup" dataset. The analysis focuses on generating synthetic dataset variations, extracting Mel-spectrogram features as PyTorch tensors, and constructing a Convolutional Recurrent Neural Network (CRNN) to understand both acoustic patterns and rhythm over time.

Question-Wise Analysis

Q1. You ran the generate_synthetic_dataset script with samples_per_genre=50. If the script executed successfully across all 10 target genres, exactly how many .wav files should now exist in your /kaggle/working/synthetic_mashups/train/ directory tree?
Approach: The script iterates through a predefined list of 10 genres, generating the specified number of sample mashups per genre.
Answer: 500

Q2. Load any of your newly generated .wav files using torchaudio.load(). Given our configuration (target_sr=22050, duration=30), what is the exact tensor shape of the resulting waveform?
Approach: The shape of a waveform tensor loaded via torchaudio is structured as *(channels, total_samples)*. The total samples are calculated by multiplying the target sample rate (22050) by the duration in seconds (30).
Answer: (2, 661500)

Q3. Using the extract_and_save_features script with n_fft=2048, hop_length=512, and n_mels=128, you converted the waveforms into .pt files. If you load one of these pre-computed tensors using torch.load(), what is its exact dimension shape?
Approach: The MelSpectrogram transformation yields a 3D tensor representing *(channels, n_mels, time_frames)*. The time frames depend on the sequence length divided by the hop length.
Answer: (2, 128, 1292)

Build Your Model (The CRNN)

For the following questions, a CRNN architecture was designed combining a 2-block Convolutional Neural Network (CNN) backbone for spatial feature extraction (frequencies) and a 1-layer Bidirectional LSTM for temporal sequence tracking (rhythm).

Q4. Inside the CRNN model's forward pass, the data moves from the CNN backbone to the LSTM. If your input batch has a size of 32, what is the exact shape of the tensor immediately after the second MaxPool2d(2) layer, right before the .permute() operation?
Approach: Tracked the tensor dimensions through two blocks of Conv2D and MaxPool2d(2). The initial 128 Mel bins and 1292 time frames are halved twice by the pooling operations, resulting in 32 Mels and 323 time frames, with 64 channels from the final convolution.
Answer: (32, 64, 32, 323)

Q5. Your model uses a Bidirectional LSTM with input_size=2048 and hidden_size=64. Write a small snippet using sum(p.numel() for p in model.lstm.parameters() if p.requires_grad) to calculate the exact number of trainable parameters in the LSTM layer alone. What is that integer value?
Approach: Evaluated the parameter count formula for a Bidirectional LSTM, factoring in the weights and biases for the input, forget, cell, and output gates across both forward and backward directions.
Answer: 1082368

Q6. When applying a 2D CNN to a Mel-spectrogram, translation invariance along the X-axis (Time) is highly beneficial, as a guitar solo at 0:15 is the same as a guitar solo at 0:25. However, why is translation invariance along the Y-axis (Mel-Frequency Bins) conceptually problematic for music?
Approach: Analyzed the physical properties represented by the axes of a spectrogram in the context of audio classification.
Answer: Moving a pattern up the Y-axis fundamentally changes its pitch and instrument identity (e.g., shifting a low bassline up 80 bins turns it into a high-pitched synth).